

Overview

I am using Visual Studio Code on MacOS to run my code, I am using Python 3.9.6. You may have to install matplotlib if the code fails.

To compile and run my code access main.py and run without debugging within the terminal you will see options to select for instance type 1 and click enter will run the frequency function and print to the terminal the frequency of each letter of the text, “MasqueOfTheRedDeath”, and so on just type the action number you want to run and it will run.

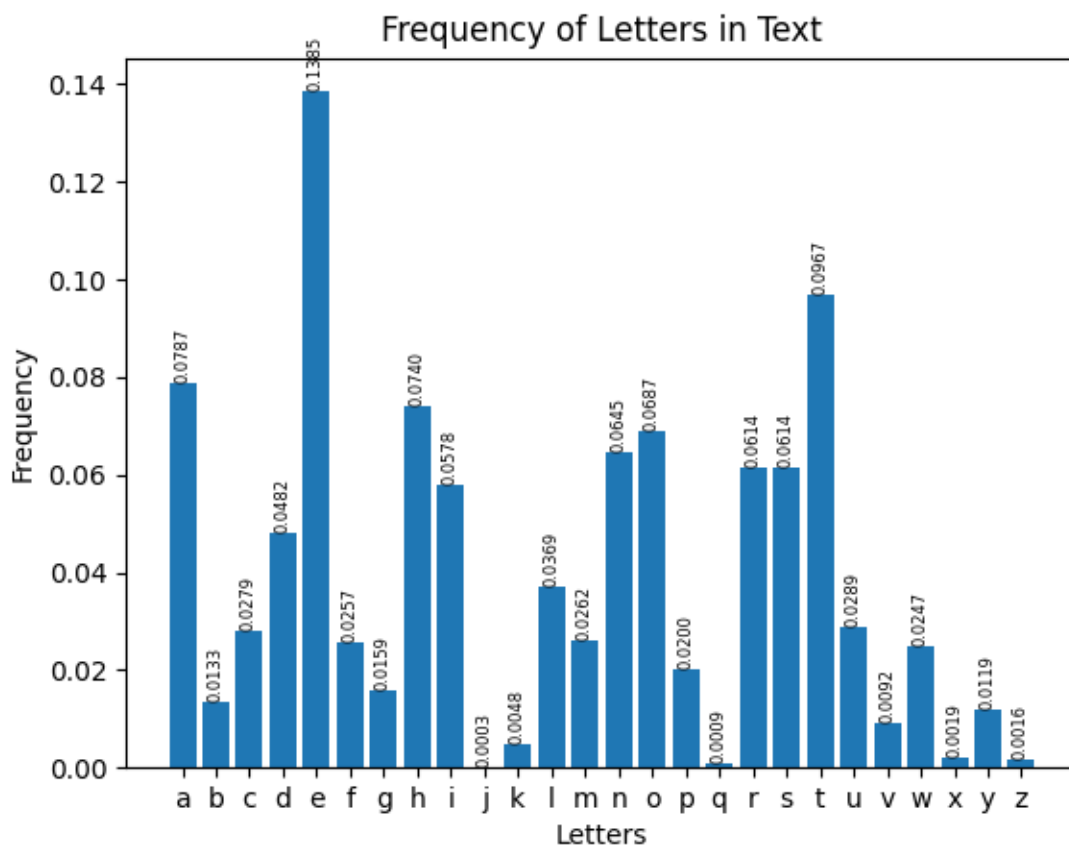
Task 1

1. Run Instructions: Run the code and select ‘1’ for frequency, ‘2’ for Index of Coincidence, ‘3’ for Frequency Bar Graph, and ‘8’ to exit.
2. The text source we are using is MasqueOfTheRedDeath which has 13,760 characters.
3. The index of coincidence produced for my text is 0.0682 which I believe is relatively normal compared to our in-class examples as this text has a more distribution of letters a-z allowing the Index of Coincidence to be smaller whereas our examples are concentrated on 100 letters with very tight distributions which showcases a much higher index of coincidence. I also believe my IC is very similar the IC of the English text provided in class was ~ 0.065 and mine is ~ 0.0682 which is very similar.
4. Screenshots:

Frequencies of letters in the text “Masque of the Red Death”:

```
a: 0.078674
b: 0.013294
c: 0.027864
d: 0.048170
e: 0.138499
f: 0.025678
g: 0.015935
h: 0.074030
i: 0.057822
j: 0.000273
k: 0.004826
l: 0.036879
m: 0.026225
n: 0.064469
o: 0.068749
p: 0.020033
q: 0.000911
r: 0.061373
s: 0.061373
t: 0.096704
u: 0.028865
v: 0.009197
w: 0.024677
x: 0.001912
y: 0.011929
z: 0.001639
```

Frequency Bar Graph Distribution:



Index of Coincidence Calculation:

```
Select what to run:
1. Frequency - print the frequency of each character in the text
2. Index of Coincidence - calculate the index of coincidence
3. Frequency Bar Graph - print a bar graph of the frequency of each character in the text
4. Quit
Enter choice (1, 2, 3, or 4): 2
Index of Coincidence: 0.06821371614363254
```

Task 2

To run the encryption and decryption functions run the code and type '4' for encryption where you will enter a message of 100 lowercase characters to ensure no spaces and the encryption key between 2-20 characters. In which it will print out the message in its uppercased encryption format. To run the decryption function, enter '5' and provide the encrypted message you would like to decrypt and the key for decryption. To exit the code type '8' to end the code.

Key String: google

Message:

thequickbrownfoxjumpsoverthelazydogandthenrunsacrossthefieldwhilebrightstarsshineabovethe quietnights

Encrypted Text:

ZVSWFMIYPXZATTCDUYSDGUGIXHVKWEFMRURETRHNPXIBYLGXCGYELKTW
KWHCVWRPFXWUNEWZOFYDLOBSGMSBSHNPUAWSZYMMVHY

Decrypted Text:

thequickbrownfoxjumpsoverthelazydogandthenrunsacrossthefieldwhilebrightstarsshineabovethe quietnights

Screenshot of the Encryption Function:

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Select what to run:
1. Frequency - print the frequency of each character in the text
2. Index of Coincidence - calculate the index of coincidence
3. Frequency Bar Graph - print a bar graph of the frequency of each character in the text
4. To Encrypt a text - encrypt a text using a key
5. To Decrypt a text - decrypt a text using a key
6. Quit
Enter choice (1, 2, 3, 4, 5, or 6): 4

Enter a text to encrypt 100 lowercase characters long:
thequickbrownfoxjumpsoverthelazydogandthenrunsacrossthefieldwhilebrightstarsshineabovethequietnights

Enter a key to encrypt the text between 2-20 characters long:
google
Encrypted Text: ZVSWFMIYPXZATTCDUYSDGUGIXHVKWEFMRURETRHNPXIBYLGXCGYELKTWKWHCVWRPFXWUNEWZOFYDLOBSGMSBSHNPUAWSZYMMVHY
```

Screenshot of the Decryption Function:

```
Select what to run:
1. Frequency - print the frequency of each character in the text
2. Index of Coincidence - calculate the index of coincidence
3. Frequency Bar Graph - print a bar graph of the frequency of each character in the text
4. To Encrypt a text - encrypt a text using a key
5. To Decrypt a text - decrypt a text using a key
6. Quit
Enter choice (1, 2, 3, 4, 5, or 6): 5

Enter a text to decrypt 100 Upper case characters long:
ZVSWFMIYPXZATTCDUYSDGUGIXHVKWEFMRURETRHNPXIBYLGXCGYELKTWKWHCVWRPFXWUNEWZOFYDLOBSGMSBSHNPUAWSZYMMVHY

Enter a key to decrypt the text between 2-20 characters long:
google
Decrypted Text: thequickbrownfoxjumpsoverthelazydogandthenrunsacrossthefieldwhilebrightstarsshineabovethequietnights
```

Task 3

How to run the attack function:

To run the attack function, please run the code. From the menu that pops up select option ‘7’, by typing 7 into the terminal. The program first takes the 10,000 characters from the text Masque of the Red Death and encrypts it using Vigenere cipher and the key “google”. The attack function then tests possible key lengths from 1-20; we separate the ciphertext into columns and calculate the average index of coincidence for each possible key length. After finding the most likely key length, the program analyzes each column to determine the most likely character of the key.

Key String: google

Plaintext:

the masque of the red death by edgar allan poe the red death had long devastated the country no pestilence had ever been so fatal or so hideous blood was its avatar and its seal the redness and the horror of blood there were sharp pains and sudden dizziness and then profuse bleeding at the pores with dissolution the scarlet stains upon the body and especially upon the face of the victim were the pest ban which shut him out from the aid and from the sympathy of his fellow men and the whole seizure progress and termination of the disease were the incidents of half an hour but the prince prospero was happy and dauntless and sagacious when his dominions were half depopulated he summoned to his presence a thousand hale and light hearted friends from among the knights and dames of his court and with these retired to the deep seclusion of one of his castles at abbey this was an extensive and magnificent structure the creation of the prince's own eccentricity and august taste a strong and lofty wall girdled it in this wall had gates of iron the courtiers having entered brought furnaces and massy hammers and welded the bolts they resolved to leave means neither of ingress nor egress to the sudden impulses of despair or of frenzy from within the abbey was amply provisioned with such precautions the courtiers might bid defiance to contagion the external world could take care of itself in the meantime it was folly to grieve or to think the prince had provided all the appliances of pleasure there were buffoons there were improvisators there were ballet dancers there were musicians there was beauty there was wine all these and security were within without was the red death it was toward the close of the fifth or sixth month of his seclusion and while the pestilence raged most furiously abroad that the prince prospero entertained his thousand friends at a masked ball of the most unusual magnificence it was a voluptuous scene that masquerade but first let me tell of the rooms in which it was held these were seven an imperial suite in many palaces however such suites form a long and straight vista while the folding doors slide back nearly to the walls on either hand that the view of the whole extent is scarcely impeded here the case was very different as might have been expected from the duke's love of the bizarre the apartments were so irregularly disposed that the vision embraced but little more than one at a time there was a sharp turn at every twenty or thirty yards and at each turn a novel effect to the right and left in the middle of each wall a tall and narrow gothic window looked out upon a close door or corridor which pursued the winding of the suite these windows were of stained glass whose colour varied in accordance with the prevailing hue of the decorations of the chamber into which it opened that at the eastern extremity was hung for example in blue and vividly blue were its windows the second chamber was purple in its ornaments and tapestries and here the panes were purple the third was green throughout and so were the casements the fourth was furnished and lighted with orange the fifth with white the sixth with violet the seventh apartment was closely shrouded in black velvet tapestries that hung all over the ceiling and down the walls falling in heavy folds upon a carpet of the same material and hue but in this chamber only the colour of the windows failed to correspond with the decorations the panes here were scarlet a deep blood colour now in no one of these seven apartments was there any lamp or candle a burning amid the profusion of golden ornaments that lay scattered to and fro or depended from the roof there was no light of any kind emanating from lamp or candle within the suite of chambers but in the corridor that followed the suite there stood opposite to each window a heavy tripod bearing a brazier of fire that projected its rays through the tinted glass and so glaringly illumined the room and thus were produced a multitude of gaudy and fantastic appearances but in the western or black chamber the effect of the fire light that streamed upon the dark hanging through the blood tinted panes was ghastly in the extreme and produced so wild a look upon the countenances of those who entered

that there were few of the company bold enough to set foot within its precincts at all it was in this apartment alone that there stood against the western wall a gigantic clock of ebony its pendulum swung to and fro with a dull heavy monotonous clang and when the minute hand made the circuit of the face and the hour was to be stricken there came from the brazen lung of the clock a sound which was clear and loud and deep and exceedingly musical but of so peculiar a note and emphasis that at each lapse of an hour the musicians of the orchestra were constrained to pause momentarily in their performance to hark to the sound and thus the waltzers perforce ceased their revolutions and there was a brief disconcert of the whole gay company and while the chimes of the clock yet rang it was observed that the giddiest grew pale and the more aged and sedate passed their hands over their brows as if in confused reverie or meditation but when the echoes had fully ceased a slight laugh that once pervaded the assembly the musicians looked at each other and smiled as if at their own nervousness and folly and made whispering vows each to the other that the next chiming of the clock should produce in them no similar emotion and then after the lapse of sixty minutes which embraced three thousand and six hundred seconds of the time that flies there came yet another chiming of the clock and then were the same disconcert and tremulousness and meditation as before but in spite of these things it was a gay and magnificent revel the tastes of the duke were peculiar he had a fine eye for colours and effects he disregarded the decorum of mere fashion his plans were bold and fiery and his conceptions glowed with barbaric lustre there are some who would have thought him mad his followers felt that he was not it was necessary to hear and see and touch him to be sure that he was not he had directed in great part the movable embellishments of these seven chambers upon occasion of this great feast and it was his own guiding taste which had given character to the masqueraders be sure they were grotesque there were much glare and glitter and piquancy and phantasm much of what has been since seen in hernani there were arabesque figures with unsuited limbs and appointments there were delirious fancies such as the madman fashion there were much of the beautiful much of the wanton much of the bizarre something of the terrible and not a little of that which might have excited disgust to and fro in these seven chambers there stalked in fact a multitude of dreams and these the dreams writhed in and about taking hue from the rooms and causing the wild music of the orchestra to seem as the echo of their steps and anon there struck the ebony clock which stands in the hall of the velvet and then for a moment all is still and all is silent save the voice of the clock the dreams are stiff frozen as they stand but the echoes of the chime die away they have endured but an instant and a light half subdued laughter floats after them as they depart and now again the music swells and the dreams live and writheth to and from more merrily than ever taking hue from the many tinted windows through which stream the rays from the tripods but to the chamber which lies most westwardly of these even there are now none of the maskers who venture forth the night is waning away and there flows a ruddier light through the blood coloured panes and the blackness of the sable drapery appals and to him whose foot falls upon the sable carpet there comes from the near clock of ebony a muffled peal more solemnly emphatic than any which reaches the ears who indulged in the more remote gaieties of the other apartments but these other apartments were densely crowded and in them beat feverishly the heart of life and there welled whirls of glee upon until at length there recommenced the sounding of midnight upon the clock and then the music ceased as she has told and the evolution of the waltzers were quieted and there was an uneasy cessation of all things as before but now there were twelve strokes to be sounded by the bell of the clock and thus it happened perchance that more of thought crept with more of time into the meditation of the thoughtful among those who revelled and thus too it happened perchance that before the last echoes of the last chime had utterly sunk into silence the

ere were many individuals in the crowd who had found leisure to become aware of the presence of a masked figure which had arrested the attention of no single individual before and the rumour of this new presence having spread itself whisperingly around there arose at length from the whole company a buzz or murmur expressive of disapprobation and surprise then finally of terror of horror and of disgust in an assembly of phantasms such as I have painted it may well be supposed that no ordinary appearance could have excited such sensation in truth the masquerade licence of the night was nearly unlimited but the figure in question had outmoded her od and gone beyond the bounds of even the princes in definite decorum there are chords in the heart of the most reckless which cannot be touched without emotion even with the utterly lost to whom life and death are equally jests there are matters of which no jest can be made the whole company indeed seemed now deeply to feel that in the costume and bearing of the stranger neither wit nor propriety existed the figure was tall and gaunt and shrouded from head to foot in the habiliments of the grave the mask which concealed the visage was made so nearly to resemble the countenance of a stiffened corpse that the closest scrutiny must have had difficulty in detecting the cheat and yet all this might have been endured if not approved by the mad revellers around but the mummer had gone so far as to assume the type of the red death his vesture was dabbled in blood and his broad brow with all the features of the face was besprinkled with the scarlet horror when the eyes of the prince prospero fell upon this spectral image which with a slow and solemn movement as if more fully to sustain its role stalked to and fro among the waltzers she was seen to be convulsed in the first moment with a strong shudder either of terror or distaste but in the next his brow reddened with rage who dares he demanded hoarsely of the courtiers who stood near him who dares insult us with this blasphemous mockery seize him and unmask him that we may know whom we have to hang at sunrise from the battlements it was in the east tower or blue chamber in which stood the prince prospero as he uttered these words they rang throughout the even rooms loudly and clearly for the prince was a bold and robust man and the music had become hushed at the waving of his hand it was in the blue room where stood the prince with a group of pale courtiers by his side at first as he spoke there was a slight rushing movement of this group in the direction of the intruder who at the moment was also near at hand and now with deliberate and stately step made closer approach to the speaker but from a certain nameless awe with which the mad assumptions of the mummer had inspired the whole party there were found none who put forth hand to seize him so that unimpeded he passed within a yard of the prince's person and while the vast assembly as if with one impulse shrank from the centre of the room to the walls he made his way uninterruptedly but with the same solemn and measured step which had distinguished him from the first through the blue chamber to the purple through the purple to the green through the green to the orange through this again to the white and even thence to the violet ere a decided movement had been made to arrest him it was then however that the prince prospero maddening with rage and the shame of his own momentary cowardice rushed hurriedly through the six chambers while none followed him on account of a deadly terror that had seized

Ciphertext:

ZVSSLWWISUQXNSFKOHKOHNMCKRUGCERZOTASKHVKCIJRSGELNORRZRMRSB
LWZOHKOXNSQUFRZFMTZTKGHOWITQSNLHKJSXMIKBGUQEZOZUCWUVWJPSA
GPRZSJKOYTXYOJGEEXOBJTXYGSGWXNSFKORKGGGYHZVSNZVXCFUQFRCCJEL
KFSCPVKGVGCTVOWTDETRGAOHKBROKDOBSYDETRHNPRVFCLFWKPZKPHOBU
GEXNSDUCIYKWZSHOGGUWYZWCTELKGQGCPKHGZLMTGIVZRZVSHZHEOBJPW

VSQOLPRMIVZRZVSLLGKCTZSIBWQZTQCSFKELKDSYEFGBKNTGNGVAELOACAE
JXCAZSIGWRGYHLFCSELKGMSAEZVMUQLOGTKWPUKAKYETRHNPNANCZKDIONI
XPTXCUXPWYOBJEIXAWTLXOCBUQXNSRODIGGSCPVKHVKTIRWKYXYCTNLPL
OBNZYXPIZELKDFOYGKDFUDTKFCCLWNODVJETRRGFRZZSYDETRGGREIWCADA
NSBNTWJCAOYMUBGCPVKVORQHKDCVFPGHSJSIYIASZRKRHUSMYDFKDITQSGE
LUIGGYHNOZKLRJZWMSXNSOXEIJTFOPRJGTZXZQGACTRXNSYTTKNHGGYHJOAK
DSLWVYNsafhgyHCWHNELKGSXPXOFSJESZVSJPIVGSIWYYWCTZJUBSUQLOGQ
GDXXZZGEIJOPHPCYHVODAGGOTPBZSBYTKOBJXEMBWLTKGBHYEVAQHACIZ
VSICIGHWUYSLHVKAVOBQKDSCBSINITHFONCKHOARYYHHGDXXKOGZCSTUOTO
PUTHEHERZUOCHRSROEMTHVODAGZZNLHMOHKDSLWFUYXNSQUFVZWSXDLGJ
WTRITHSXPHHFCARLZTIXYEISGGYHSOGYJLGAACWGBRCPPIJRZSIHCZZDXNS
MXPWUZJKOXUZSGGISSOTDRKWHNPVUTWTRVKGGTZVKUFKDWZCHNPWARRK
YMSDIRDIYCTJPWVOWXZVUTTXPRFMTXZQCWHNTRZVSGMFKMKGDESDZEAVU
JWYTSTSRCTXNGIISTXSQGF XOBYELKQCACXOSFYXMMVHHTHJSTOLRISHUNST
HOMTSTHVKPBZSFTLPCCFROGUIZJEEQSQGCIUTWZDIRTWTELKASGYXOASOEAG
GTUWPEHCMCMKJSUCXUHVOYOZVSVCMTQSNLHVFCBTHKRORWXNSOVAPOOB
IPWUTDRPEYIFKELKFSCPVKPILQSUBGZSIXSKKCIADXXZZOGOZZVOHVKCICSFK
MERZSZOETQSXDXNSFKHIXSAADMIWOTDXNSFKHEYPSGF XEHVKCICOGCTRKOZ
RELKSGYHYSQACMZMKKCICWHNTRCWHNZYZKOYELKFSJOIGHVOEAGGHUHE
XRGZSIIZCYP SLHVKQMLHVUCWOLHNXSTHVUQLOGGKNPAGWUYETRKNTPKHV
KAIYHWRPRISFGRIJACYEJAFWUFWRMOHCSGRHNLXZVSVCMTQSVCSYDSXZITH
SXEEOBSJSMYHVUFWGBRLCMKBRYLXGAOYVIJPORWSLHVKXSYHITFWAOZSLK
TWTONITQSOEAGGOBZPADHAZYYGQKYIZVOZXEYEIKCEJSPAEJOFGZWIZASZPP
RCTZSIXCCSDMTKVONLOHKGDLKZRZSIYSKKCIYSJKYETWAVPVOOZYFMZSWTX
ETMDGWEISGNZAKJSXDYIVGATXKGTUCQGZCTRETRGZCEOUVZGMYHOCMSMRSH
NPJUZROYKJCCXDWRWRKMEIYBKLVRMHUELKKORWWUBSOELKFVGYHYCHNL
XZVSBTICCTZSICVCRPIDHSTEMYGQGCGKZMOXTKRSJSIXSHNPGGGSC LWBSFEO
MLTSXPRZOGSTKNHVGGIHSSTPBVSQZPHLFCSELKRIQPWRCJKZJZVSHTDGGFFKEL
KODGCXSSBZDAKFSYZMXFSMFPGFZEOMYDCYPHZVOZELKJWYTSTSAHCEISRHF
XRWHZWISCFKELGBCTPEZOHXIZVSPAGGOYSEXDHACRGHSBPVEHKKYXECF
ZSMXHMELVJGOTOEZSOISXAFBGYSBSZKQJKQHZZXNSFORLZOBJWILHWTELKA
WJOPKCTKLGNKORWEZOZRLRJBXCSCUCZSMIKWTOSCZCUVIJCIZFTUBOIWSYS
RIZVXWRUCANWQNA YXGIKXNSKOYHOBUYZJZVSYFMZSHNPWKKWTOSC GKK
CIUTGZLMTSRMWEYGKNZWKQCRZYXJOXTIJWBGNGUFRGYGKKWZSXNSDXPZG
WZOYKNISUQXNSRKN SXOH ZRYCTZSIIVOSMIXWBZZANWQNTXUDSTPHZVOZL
XZVSKLWZSFTPBZFSSTXEKOYSYTUTUCIDOAVWIOBPRFIGBRBTZORZEMPASKKC
IOHGCTRJCKYELKGSIZRJQVGXFKFKGDTAFDRPMTWHYZVTOAKYXYOBJEEVSGZ
CMKGOTOLKFSZSIVOBKDAKFSVFVVZSZSIZVWXOAGGUXPITHVXZYMVCAEETR
GUHIXSHNPGGGSSPRZGHNPJUIFZSAGGTACROGVKOETRZORLZSRCTXNCFGYKK
HVKQMLHVCTXNKVOEIZVSYTBZVKOELBWCRPXZVSY PZKBHNLTFHSPRZKOYN

PUGSRJWNFCAOIJBHWEIYJKWZKHHGAIYHFOPWZVOZSYTUORWSBSFZSIISWR
TRMOBJOSCBHNPAGZZYQERZWTRMTVSGGCLCZJDYVCBGNEXDSZZJZVSYLQKA
OZPVOOZGYHNISHFXOBHNTWIVOSMIXCBRJXNSQUWSAFCLELKKWTOSCGTGTP
KRHUNSXFSYASTRKOELZVSJPGUFOZTSTGHNPGBSYSIXSKKCIYQOXWIZORKPT
HZCUOGUZCACRUKWTYSUBSUQXNSGKGITODGCXSSBZDAGGHNPVKOBESWEDC
XNETRSRLFXIAGXMJHVKAUTIYTSTCTMZPJSBUCRGASTEWZVOZWEEGQGEXKF
SJESGBRLCSUFRKAITRSJQVUAHNPVUCTZSIXSKGDRUZWMSXUTOTJOBRKXETO
HOYKLFCSWESDCXNETRZKHMZVWTELKGIOEIUTQNLQHSFYMYZWBZSIICFXTHU
FGZSEZTCRWSCSRZSIYIWZPXNSFKDXUCRUATUGWZPXUSOISAABRUHENSBJX
XWDUOFKOFKYKPGFKMKFCLQMXSHNLXVFCPPGZSROEWXOMYELXCIMSXNS
HOYXKRURLWYOBJDSMZOXTRMZMOWPAAWTPHZVSXZSSOBJELAGKKCIVFCJFG
KROSPZWHAOIUTUGFHEOBJQETHOYEMIODVPEXOBIPWHIHOYXNSKKDXKFBUC
FROQQNLGAPKCNSSLQIIHCLELKTWXPPOUVZELGHGZCIGASJFTUBHNPFGFYNL
RMWBMDXNFCARLZVSHWSURHOYXKRDGYIYKOYRLGGHRJMTHTVKBZFSSETR
DXZHAQSDSCWZJLPUCYAASTHVKNABHKEYTQSYZJZVCYPANCSTEIXSRZSEZH
VKCICSFKQICCTZSIICAVLREPCROITCIMSXUGSZQSUHKOELOBWZDTXSQOYGG
OZLPRWHCLWOBHNTWGDQXEQKBHGWUHVGEHNSFKDXUCRGREOBGZELKKS
YEIXBKGPWGUWMLRZWQIWSIYCLPFUBMOEWSBJFPAAGCFRMHCGYHLFCCTX
NORAWPNBQJUBCZZRUIGIWETUOTOANSBZSISWBAEINOBXJESHNPQOFQATX
UTHNPJGQSGYHZVSNZYXKOYESHSGZCMIYSTELKFSILQKTFUXXNSPXLDBZAY
KYCTZSIIZCIVEYCITOANWQNHEYQZKLVGBRRZYJOBJOIKDOTOIDQSKOMTUZEX
YYWQGWFAHCLDSVSQAWMGFOTZXKOBJPQVVOYTWZVOZLXKOQNWEVGSUQE
TVCACXNSAADMIWOTDSLHVKZVIVSYEVGKSXPBGZCEOBSJESVOIYPQUASTE
EXWZETRZVSOCTKFTUCQGBQKESNOFQPRZCHNPWUIBJLRJHVADXNSKGWXSFS
YAIXTCXNIISOYPHZVSOCIBCZAEMUBGGYHZVXSPAGGOHCMKTRODUBQKCKXU
THNPANCZKREEQCSAETMOTOANWZKELKQVOXIYCTZSIIZCIVCKHFGYKOHKGDS
HGSXGIJHVGEXNSUOOHOSGZRVKKGWIGBRZSISCFKLKKROTOWKROZPTGGGK
OXNSWXSETRGUGIXHVKTVHFCCDEYWTOYGUBTADIJFSBPVOSCXXIJWHGEMUB
PAEANSBZSIKQVUPWNORLFPRMQKLWKRORTKNHZGFKNHSXLXUBQKAIXJOJPH
ZVSGDWKAPRJXNSAADMIWOTDPUCYKOEZSOISSZVSXLRJGAOWIJOGOQEZHVKT
VUKBTPVBCIYYIYGOTOJUZZELRJAOJPANWGVPOBUBZAYSOISXUHVKZXNSFZS
EZHVKYIDHQNTQOBUUQXNSQRZGQGVUFPJDFUOYISWTELKABUDMSWZGCISCH
OZRGRZSITOTZPVZVSRLTYSCLDMDHMSTRAHSYHLOQVKXFXOQKELXSSZSSAG
OTOETRGOILABRXPYHSQUYHYCTZSIZWAKELGHTRTIYHVKCIIIOAKJIZOBUELKFQ
NTQOBUUQXNSQRZGQOBJELKBKKCIZVSYLQKRWYNSTQSXEETRHXPQAZCADRK
GGGYHSSROEEZWCTLWHSTUCIHIOYVWVHKZJZVSYPNWBMDMZKOYLKGM
TOQGUBOQMISBZCIBSZSIZOGZPWUTHNPHAYSCPVKDSIFPOOFNPLGROLTRKSM
KQSQCRZYXGOTOILTSIEWNSRODVKUOXOIJHVKOIICFGZJSSFKQEYVWUYLOGD
RLRYKSXPFUZRGYHLWSXJETRVODGUBQKAXOCBYRPUKSJHMZVPGCFGFWIY
YHFKELKFSGCIYCAKHLUKCAWHNOJKELUIUNELOAAGOLOGTUWPUKSXDJKZH

SEZVSCLWTCHOEAGGBKNYGOXJXUVSGCETRGPETRHUFGNVWSESHSGACIZV
OZSICOGTZXNSVGOHOFSEIEJWBMCIGHDGCXZVSSZZGPZKPQHSZRTWNASTEWUT
HNPWKJSTNLGAPKCWADCTZGIOGOZRUTHNTWMFSGEJZSOTOMZKOYSMYCKTR
YORWTRXGGHKLHOQVNLHMWJKYGN OFGNXKFHUELKA OYBYKFOJPVYPSYFVK
HVKJAKFSMCSZSGWFIZVSXPAKFSSFGNUZGCIGBRMWMZHSLRJDWWFETQMGY
HVVOTE EYAAANLUTKNLXNOGHPITGWTNIYSSTTRNSFTLROHVKCICSFKLVGPSY
BYKTWMFVKGKOELABGATXKRZOXFYOBLTVCWTEQKBHYELKFSCPVKRSRTVO
CIYQETQWKDWAQVGD XNSAGOQGBTGDLOCBYELKFSCPVKAISSSLHVKMIGIHOQ
YRAISSLHVKHETHCTXYIVCLELKPWFLVXS GUXIZVWTRSLHVKEIXFWHWIGBRTZ
XGZWZEPKCTZSEZKVONLSWUNELGJSKIGOHSJOMYUIYEXUOBJQVUWBZSIYSJKY
GNOAHPVYHVKCIYHORVIJWBLLGZOAAWXOHIJPSLRFKLQYOBJELKGSZSIJFSGX
WCFWZSIJWBGYHGPCAEXGYWTRLASTXZQZVSXZSSGOTOGGIGOYKZVSCTPJAIY
TGUTHNPSXQVKDXXOHUDIKA OYELKSQNZSLHVKT VYHSVDETROTZRZVSXPWZF
WQPWZVSKMSTMQRZGQKVONLYHOTOWOBHNPLGZZUQXNSJKWZKHOTOXNSBL
ZVGACSPRZOZRTWYHWRWETRORWMYGWRPRZGOBPXNSJUTGKCTZSIIZCIVXNS
RXPESGOXPWZWTLQVUNSTLWZVSEDXGBRHFZXVSKNLUSGUQXNSQNTQKRWKL
AGMHNPCNOJKPRJIFKOF AHOTTRYHOTEETRORTKNHVGWJYIPJFIJZOARLZSFLWS
GHGGQXKFHNPQGGHNPCJSDGCXGBRTZAGUOOYXNSAADMIGKKWPYOBJELKRF
KLQYZWBPETRKXTXNSHULRJTFUXSXS AKCVOZMZSETSJKCXGYWTRLASTXZQZ
VSSLREHWTEIJKWTOSCGHNCSAUVC SMIVGZCIGAHPVGMGLCSSHVKEVODCJDF
AHHUELKQVGXFKFKNTGNZWKDQU GHCPWZKXOXOPECTZSIYSJKYXNSFKLVKBC
CYSTSCLELKA OYVIXGKNZZKBHACILCFZSITWUNEMYKOTTRMOKGJETRHNPVKT
ZUHWGFIJOMKFZORLZHVXZYMVHNPFRC CJNSRCIXPHVOBKDETRHNPFRQQYIY
GCLELKG OHWIJFOVPVEODVLPYOBJESNWACSSYSTUZXL OZRDYVCBZSIYOPRPG
GFDKEXNSFKNSSSGLCSSHVKYIGFQRZGQCTKMSTMOSFJLZSJAIGZAUCIYCZKXR
RMSSALGHWIELGBOTJANWQNCIGQVKDXNSWXP EXGKNZMTRIRRIJBZSISCFKC
ISCHKREOSHOPWUTHNPSZVSXLTGFHSPRZGPAEXNSGKZXNSFGAEXHAKYXYKXS
PHKBGKWCIFCCOIJOB JTRZVSSMIGHTKGIXWGNWCZVSNPEXHCLWMLSOTOXNSF
KGIRKSTEANWFRTRMZMUY YTHWRLXRSBMELZVSXPGUAAKYGKRHNPWUIBJTR
MCTSTHTWUNEYVCBZSIIZCIVETRHNPRZVSSFWOQQLWKROYTLGJSZZPJOBJEL
KSJUWYZWCTDSLHVKHERHNKCWC SFKBYOSHKOETRHNPKOYL RABSGDCISG
Y LXOCBUQERZHNTRMGOYMILCFKMYZBCCELKFSCPVKHKKWZKGHXZOKGHUM
IYCITOIJPMZSIHSZRZJZVSIWSIYOTOXNIGOELGDDKYIJD SXSEVGHNLXSCFKZJZV
CARLZQFKAXCWHNXSXS CLEMSSWTESZVSSPHOHOZTSTGCLELKHVUFKNHTAWE
SCBMELUGSCSSXSJKWPKROTOXNIGZZSOHV GATKBSJAIXVOVDXNOHHPJUFZSI
ROGZPGNCSYZJZVSRLWZQVOXINORAEXKFZEDYTYWTESYWZKYGKHVKCICSFK
XETMWTOMBWRALPYWBZSIIFCCOANCVG OJUIBJWIOGIXPXUPSIZQKOKGCIUTHN
PTXSGKYGKCTGXEYYSJQMMIFKHLOQVNLHGFFKDXKRHNPEZHSTEMUBCLYSY
WBMWIOBROGMJIORMILCFKL RJHVKCYSCIXZJZVWYYICDFKDITQSNLZOBUYAV
KOROEWKZTC SMYDSXTRMZMGCSABRZSIXSOXZWKOHRPRMHVLCSSHVKHLUZS

IZQVOBELFANNUCQAFACIDDFKDWOJSUQHOGOVAVUPOZTSTOBDYXDFODIZV
STQMTOZRJSLHSXCSXCTNZVXCFGYHUTRODKAGHOYETOGYPQHZMUQTNBZL
WSGGANLGGWNLZKDOOYXKRWZXEEKSRWFKGIVASYSRZSEZBCUCHOBXJEVD
SGCETQSIZYRRVGGIKLQOEIJGIISWKBGGEMUBWTEVAHVZSISOGWFIXORKWMIS
BIPSLHVKYMMVHCLWTSOXWCABZOXMZSRHFXZVSLTKAFSOYUASGZTSTVOJZY
ZVSXZHKRVKCSJOBJRSTSPKJSTRHNPFUIBJDSLJSJKYXNSDXTRISGOYHKTWTTXK
RSIZVAAHNPVKOFKNLUFRTYTRZVSNPEXHGUQXNSAUDXXSQQWIYGKNTGNQOT
YSZPSZZYIVSJHMZVCAEISCHOZRKJSTHMZVHNPHYZHSXWCRCGZESCVCSWMLSO
TOHKOHNLVKSEALPRMXKDXHYHVKCIGFSSLXZSFYZJCVWISRUXSYEGGBPKXEJS
HNANCZKNSSDOTJMTRSKOWKSAKORUKRKPTRMHUQIKZHNLXOBHNPUGUHAX
IGBRHPEXWBMZJZVSYEVGBUKCRKWHNPVCWHTZVVFVCVCMKHKIMYHSJELKT
WMFVKKOYEERZOTOKGIBZLRJGVXZYJSRLCSSVSGOXUTCUEMTHVKSEHWZOXI
THGUQXNSUXLZKHVKXEYYKNTGNQCTNIGZSJELKJWYLKKKOYXEJSGUYIGFZEE
SXSGKXFRSHNPGUIBZPRGBQKZJGGHOQJKBSJNSXDGKELGHHNPGRCGKDXQFA
EMTMAADXNOJKSEJRWLQMIIZZJMTRSZPGZWBMEKQVKLXGBREPXGZZSMYA
WMSXNOJKMIKBSTOYXSROQRUHOVAUVJSJMCZVSSLHXSJKWPKFGGCSABRHF
ZVSSFQSSFNLHMCBKDSLDFGXUOGYFQKHVKECVSCLELKFSJOIGHVNTWBSGZF
VKKOYOEHPZKOMTPZUZHGBRNTWHFCGOFXCKCTXNOZRELKTSGEYXSGUQXNS
TGNICOGHPWVFWTVPKRKOELZVSYNEXSZSSXFCXHLKBHNPESGUQXNSDXTRI
SDXZWVSFUQIRZIVZRZVWYDTKQHLPOAOMPANWQNHMZVOYWSCOBJSRSAT
XSBSAKYXGGWLXSXSTAWPEHCYFWZOWTTXYFCRPWZOZQPHZCOTOJXCOSZRM
HVKHERHNKCWNSKGDWKSBBZZFKQCTGYRGSJTRZVSLTVYHAUXITHKOELGGHX
ZRMGVAOHKFSOELKFCLEIXFCXZVJWGZLWZSPAEMTHVKYIDHVODFXCKXPHJSB
KAOHVXLKKKVUOEXSGNPHKAOTOIJVCGCWKZMUQXNSQUFVZWSXDANCGZZ
SJBSGCLOAKNZHGFSYTRYZZFWCWHNELOGPRLWVVSZYYACIVIXMGKTDKVV
SLRJIBSLWQVWSELGHKKXEEYBUHANCACPLGJSZZLGBUGEWABFODILFCSELKP
OZEPKASTEWOHKGDMDTHVKPEYHSXYSXPZAPGNOAHPVOBKNTGNGHUZHVSVC
MTQSVCSYDSXZEYVSAEXKFSJELKGSCZVJGHNPCXOBMELXCIMSSAHHNPWKJST
CSUAGRZYJZMGYHIZSGCPETCXELKDFOYGKKOYLFUZRGYHXCPCADXSGBGYHZV
SSFWOQVGOFKQCSPLAGVKOEZHVKHEBWBWZJNWGNLRJWHCLWOBNPFRISXZ
SSKVKCIYHCUOXNSDXTRISKOELGUFUFTUTDGWIIICIXEMKFGHJLOGGOOIGHTOC
WZOGNPWVCYKELKFSCLVGGZORLZFIYSMTUAUGISSBZZJZVWYRVUIDOYXNSR
OCIIHWUYSLHVKTRZFIJPVCVCSEXNSAUXITHKGDERGCTPEXOHNLRJOBJSCKW
ZSHKZWHPVGHSGYHYHOZPPEGHKAQGRSIWSYSFGATXCOISXUHVKDTKOYKCF
HTXZQGGQSXEEOBBGXIRSGYLAKKWZSANWQNELKAOJLWYIAVEMUBGUQXNSAA
XQKFVGOMTGDOCIJHVKHLUZSVLVZMHNPKKSPJUIBJYSTSKNZTAHTUCXNVO
TOXUGSOKINWAYZXNOHAYMSDSJPHNSDGDWKRKOELOBOELVJCTZSIVFWTNIY
DSXDSTOBHLOZSZSIBOGZLWYSAHWCGGWLHMZVCTPMSDIRDIYVFGYOLFCSEL
KQSTEVKGCLELKFCUXWZCHNPAGZZYSISORKSMYKOEFRQBHCKVADHKOPEPIZ
HMZVHNPPWGASYZPKABGYHSSOYFVKRGZPTCVWISLGRRODXOBUATWNSRNTQL

FCSELKTWDXZVFUFKNHVKMPASQNLQHSFZZXNSDACTRSHNCSAUVZSIVIFVWI
ZCHNPKXSSTELXCIMSXNSUXPITHCZSIUFOTRIZVFUFKNHVODEMOWTESZVSCSM
ZSOTOIBSBZSITQSZZXNSJOZPKHSXPEJSQOOIJACBPQKBHNLHHSSTXEJSHULVXS
GZSMSWHCLWZVSTSSCSJKCXNOHZSIVFWTNIVFCYAIXCAGOHKBWTRAOHVXLK
KOBJELKGVGXIUTVODSCBAUXITHOXJGUKOXOMISFADLKRVACVOSRRJXNFCAR
LZVSYTBIVOSMIXGKNTPKBCTPJUZZUHIJVWSZRGQQUFRZCTGOIGRZEEIXFCXEL
GHVGOWKWNK

Testing key lengths:

Key Length: 1, Average IC: 0.043558

Key Length: 2, Average IC: 0.045254

Key Length: 3, Average IC: 0.057700

Key Length: 4, Average IC: 0.045248

Key Length: 5, Average IC: 0.043491

Key Length: 6, Average IC: 0.068373

Key Length: 7, Average IC: 0.043503

Key Length: 8, Average IC: 0.045184

Key Length: 9, Average IC: 0.057876

Key Length: 10, Average IC: 0.045159

Key Length: 11, Average IC: 0.043530

Key Length: 12, Average IC: 0.068333

Key Length: 13, Average IC: 0.043621

Key Length: 14, Average IC: 0.045117

Key Length: 15, Average IC: 0.057712

Key Length: 16, Average IC: 0.045116

Key Length: 17, Average IC: 0.043767

Key Length: 18, Average IC: 0.068490

Key Length: 19, Average IC: 0.043481

Key Length: 20, Average IC: 0.045277

Best Key Length: 6

Best Average IC: 0.068373

Column 1: Best Shift: 6, Score: 0.040764, Key Character: g

Column 2: Best Shift: 14, Score: 0.035685, Key Character: o

Column 3: Best Shift: 14, Score: 0.041176, Key Character: o

Column 4: Best Shift: 6, Score: 0.029740, Key Character: g

Column 5: Best Shift: 11, Score: 0.030033, Key Character: l

Column 6: Best Shift: 4, Score: 0.045588, Key Character: e

Recovered Key: google

The attack function successfully recovered the key string correctly. The original encryption key was “google”, and the attack was able to recover it without incorrect characters. The attack was also able to determine the key length of 6. The IC results support the key-length determination since the average IC for the key length of 6 was 0.068373, which is very close to the expected English IC of approximately 0.0667. It should be noticed that other key lengths occurred, 12 and 18 which are similar because they are multiples of 6. Overall, the attack function successfully uses the Index of Coincidence to identify the key length and frequency analysis to recover the individual key characters. The recovered key matched the original encryption key exactly. I also ran into a problem trying to determine the best key length due to some values being relatively close to the $IC_{\text{English}} = 0.0667$ so I implemented chi-squared scoring to determine the best key length by decrypting every character in column where the expected frequencies are stored. A lower chi-squared score means that the decrypted column’s letter distribution is closer to IC_{English} . We keep track of the lowest scores and return the most likely character for that column. In our case, the attack and our scoring are able to produce the key “google” from the encrypted message.